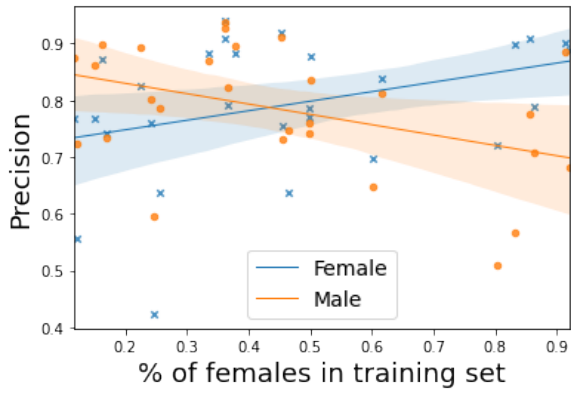
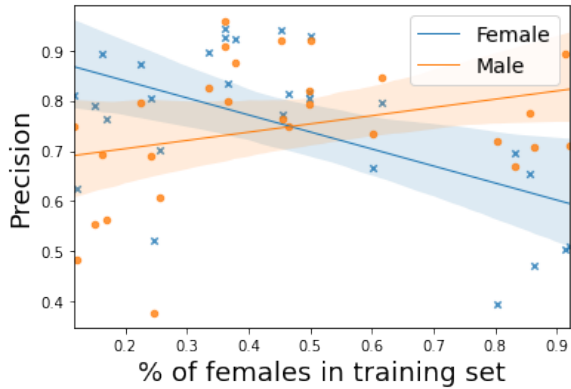


Metric	Bias in Bios (Original)	Winobias
TPR gap (P)	0.304	0.022
TPR gap (S)	0.449	0.002
FPR gap (P)	0.120	0.030
FPR gap (S)	0.046	0.008
Precision gap (P)	0.063	0.013
Precision gap (S)	0.291	0
Independence gap (S)	0.382	0.005
Separation gap (S)	0.165	0.001
Sufficiency gap (S)	0.567	0.025

Table 3: Correlations between intrinsic and extrinsic metrics. Original correlations are from [Orgad et al. \(2022\)](#), our correlations are calculated with the intrinsic metric as measured on Winobias.



(a) on an imbalanced dataset (same distribution as the training set).



(b) On a balanced (subsampled) dataset.

Figure 5: Precision of female and male examples over all professions, as a function of the percentage of females in the training set.

D Full List of Reviewed Papers for Extrinsic Metrics Measurements

Table 5 presents the papers we reviewed and the amount of extrinsic metrics measured by them.

	Females	Males
professor	45.10%	54.90%
accountant	36.73%	63.27%
journalist	49.51%	50.49%
architect	23.66%	76.34%
photographer	35.72%	64.28%
psychologist	62.07%	37.93%
teacher	60.24%	39.76%
nurse	90.84%	9.16%
attorney	38.29%	61.71%
software_engineer	15.80%	84.20%
painter	45.74%	54.26%
physician	49.37%	50.63%
chiropractor	26.31%	73.69%
personal_trainer	45.56%	54.44%
surgeon	14.82%	85.18%
filmmaker	32.94%	67.06%
dietitian	92.84%	7.16%
dentist	35.28%	64.72%
dj	14.18%	85.82%
model	82.74%	17.26%
composer	16.37%	83.63%
poet	49.05%	50.95%
comedian	21.14%	78.86%
yoga_teacher	84.51%	15.49%
interior_designer	80.77%	19.23%
pastor	24.03%	75.97%
rapper	9.69%	90.31%
paralegal	84.88%	15.12%

Table 4: Statistics of professions and genders as they appear in the Bias in Bios dataset.

Paper	Maximum # of extrinsic metrics per task
Bolukbasi et al. (2016) ; Zhang et al. (2018) Bordia and Bowman (2019) ; Ethayarajh et al. (2019) Sahlgren and Olsson (2019) ; Karve et al. (2019) Hall Maudslay et al. (2019) ; Sedoc and Ungar (2019) Kaneko and Bollegala (2019) ; Liang et al. (2020) Dev et al. (2020b) ; Shin et al. (2020) Kaneko and Bollegala (2021b)	0
Zhao et al. (2017, 2018b) Li et al. (2018) ; Elazar and Goldberg (2018) Zmigrod et al. (2019) ; Zhao et al. (2019) Kumar et al. (2020) ; Bartl et al. (2020) Sen et al. (2021)	1
Prost et al. (2019) ; Qian et al. (2019) Emami et al. (2019) ; Habash et al. (2019) Dinan et al. (2020) ; Costa-jussà and de Jorge (2020) Basta et al. (2020)	2
Park et al. (2018) ; Stafanovičs et al. (2020) Saunders and Byrne (2020) ; Saunders et al. (2020) Kaneko and Bollegala (2021a) ; Jin et al. (2021)	3
de Vassimon Manela et al. (2021)	6
Orgad et al. (2022)	10

Table 5: Papers about gender bias and the number of extrinsic metrics they measured per task. 0 means no extrinsic metrics were measured.